



12774 - Taking aim at Precision Cosmology with SN Atalanta

Cycle: 4, Proposal Category: DD

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Aadya Agrawal (PI)	University of Illinois at Urbana - Champaign
Dr. Justin Pierel (CoI) (CoPI)	Space Telescope Science Institute
Conor Larison (CoI) (CoPI)	Space Telescope Science Institute
Dr. Massimo Pascale (CoI) (CoPI)	University of California - Los Angeles
Prof. Gautham Narayan (CoI)	University of Illinois at Urbana - Champaign
Prof. Masamune Oguri (CoI)	Chiba University
Prof. Masami Ouchi (CoI)	National Astronomical Observatory of Japan (NAOJ)
Dr. Seiji Fujimoto (CoI) (CSA Member)	University of Toronto
Dr. Anton M. Koekemoer (CoI) (Contact)	Space Telescope Science Institute
Ray A. Lucas (CoI)	Space Telescope Science Institute
Prof. Marusa Bradac (CoI)	University of Ljubljana, Dept. of Physics
Dr. Gael Noirot (CoI) (CSA Member)	Space Telescope Science Institute - CSA - JWST
Christa Noel DeCoursey (CoI)	University of Arizona
Mike Engesser (CoI)	Space Telescope Science Institute
Mr. Zac George Lane (CoI)	University of Canterbury
Dr. Matthew Ryan Siebert (CoI)	Space Telescope Science Institute
Dr. Massimo Griggio (CoI)	Space Telescope Science Institute
Dr. David Coulter (CoI)	The Johns Hopkins University
Dr. Armin Rest (CoI)	Space Telescope Science Institute
Dr. Melissa Shahbandeh (CoI)	Space Telescope Science Institute
Rodrigo Angulo (CoI)	The Johns Hopkins University
Koji Shukawa (CoI)	The Johns Hopkins University
Dr. Louis-Gregory Strolger (CoI)	Space Telescope Science Institute

JWST Proposal 12774 (Created: Tuesday, June 23, 2026, 11:00:18AM Eastern Standard Time) - Overview

<i>Name</i>	<i>Institution</i>
Prof. Haojing Yan (CoI)	University of Missouri - Columbia
Bangzheng Sun (CoI)	University of Missouri - Columbia
Dr. Gabriel Brammer (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Dr. Vasily Kokorev (CoI)	University of Texas at Austin
Dr. Joseph Frederick Victor Allingham (CoI)	Ben-Gurion University of the Negev
Dr. Ori Dosovitz Fox (CoI)	Space Telescope Science Institute
Dr. Takashi Moriya (CoI)	National Astronomical Observatory of Japan (NAOJ)
Dr. Yoshihisa Asada (CoI) (CSA Member)	University of Toronto
Dr. Rachel Bezanson (CoI)	University of Pittsburgh
Prof. John Chisholm (CoI)	University of Texas at Austin
Dr. Dan Coe (CoI) (ESA Member)	Space Telescope Science Institute - ESA - JWST
Prof. Pratika Dayal (CoI) (CSA Member)	Canadian Institute for Theoretical Astrophysics
Prof. Steven L. Finkelstein (CoI)	University of Texas at Austin
Dr. Lukas Jonathan Furtak (CoI)	University of Texas at Austin
Prof. Robert M. Quimby (CoI)	San Diego State University
Dr. Fengwu Sun (CoI)	Harvard University
Prof. Tommaso L. Treu (CoI)	University of California - Los Angeles

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	NIRSpec	NIRSpec Fixed Slit Spectroscopy	(32) SMACS0723-GLSN-1
	2	NIRCam	NIRCam Imaging	(32) SMACS0723-GLSN-1

ABSTRACT

A multiply imaged supernova, SN Atalanta, has been discovered in JWST/NIRCam imaging of the galaxy cluster SMACS J0723.3-7327 — the iconic Early Release Observation deep field. Two images are currently visible at ~25x magnification, with a model-predicted time delay of 20–30 observer-frame days. Available photometry is consistent with a Type IIP SN near the end of its plateau, a phase whose sharp subsequent decline is well suited to precise time-delay measurement. However, the extremely faint host galaxy (~29.5 mag) has no spectroscopic redshift, and the SN classification remains photometric. Both are required to convert the time delay into a measurement of H0. We request 4 HST orbits and 3.3 hours of JWST time to obtain multi-epoch imaging of the post-plateau drop-off and a single NIRSpec PRISM spectrum for redshift and classification. With

existing lens models anchored by deep multi-facility coverage of this cluster, we estimate a $\sim 5\text{--}10\%$ H_0 measurement from this system alone. SN Atalanta is actively fading and will be beyond spectroscopic reach before the next JWST visibility window.

OBSERVING DESCRIPTION

We propose 4 HST orbits and 3.3 hours of JWST time to follow up SN Atalanta, a multiply imaged supernova discovered in the galaxy cluster SMACS J0723.3-7327. Three epochs of HST/WFC3-IR imaging in F110W and F160W will sample the post-plateau decline of both SN images, providing a precise photometric time delay. A single JWST visit pairs NIRCам imaging in F115W/F150W/F300M/F444W with NIRSpec fixed-slit PRISM spectroscopy. The NIRCам epoch fills the HST observability gap and matches the discovery filters, while the NIRSpec spectrum delivers a spectroscopic redshift for the faint host galaxy and a SN classification via H-alpha and other diagnostic emission lines. Together these observations enable a $5\text{--}10\%$ measurement of H_0 from one of the most well-characterized lensing clusters in existence.

Proposal 12774 - Targets - Taking aim at Precision Cosmology with SN Atalanta

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(32)	SMACS0723-GLSN-1	RA: 07 23 17.7040 (110.8237667d) Dec: -73 27 10.74 (-73.45298d) Equinox: J2000		
	Comments: Category=Star Description=[Type II supernovae]				
	(33)	SMACS0723-GLSN-2	RA: 07 23 18.0329 (110.8251371d) Dec: -73 27 7.89 (-73.45219d) Equinox: J2000		
	Comments: Category=Star Description=[Type II supernovae]				
	(34)	NIRSpec-TA	RA: 07 23 13.6422 (110.8068425d) Dec: -73 27 6.23 (-73.45173d) Equinox: J2000		
	Comments: Category=Star Description=[O stars] Extended=NO				

Proposal 12774 - Observation 1 - Taking aim at Precision Cosmology with SN Atalanta

Observation	Proposal 12774, Observation 1: NIRSpec										Tue Jun 23 16:00:18 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(NIRSpec (Obs 1)) Warning (Form): Observers are responsible for checking that target acquisition is feasible.											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(32)	SMACS0723-GLSN-1	RA: 07 23 17.7040 (110.8237667d) Dec: -73 27 10.74 (-73.45298d) Equinox: J2000									
	Comments: Category=Star Description=[Type II supernovae]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	34 NIRSpec-TA	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628		
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				FULL			
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	5						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	S200A1	NRSIRS2	9	1	1	NONE	5	5	3355.445	

Proposal 12774 - Observation 1 - Taking aim at Precision Cosmology with SN Atalanta

Special Requirements	Before Date 07-JUL-2026:00:00:00 Group Observations 1, 2, Non-interruptible
----------------------	---

Proposal 12774 - Observation 2 - Taking aim at Precision Cosmology with SN Atalanta

Observation	Proposal 12774, Observation 2: NIRCam Diagnostic Status: Warning Observing Template: NIRCam Imaging									Tue Jun 23 16:00:18 GMT 2026
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(32)	SMACS0723-GLSN-1	RA: 07 23 17.7040 (110.8237667d) Dec: -73 27 10.74 (-73.45298d) Equinox: J2000 Comments: Category=Star Description=[Type II supernovae]							
Template	Module		Subarray				Target Placement			
	ALL		FULL				Module B center (small extended source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	NONE				SMALL-GRID-DITHER				4
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F115W	F444W	BRIGHT1	10	1	4	4	815.995	
	2	F150W	F300M	BRIGHT1	10	1	4	4	815.995	
Special Requirements	Fiducial Point Override NRCBS_FULL Group Observations 1, 2, Non-interruptible									